

University of Asia Pacific

Department of Computer Science and Engineering

CSE 437: Robotics Lab

Project Final Report

**Project Title : Ultra High-Speed Autonomous Line-Following Robot
Precision 6-Channel IR Sensor Array and Advanced PID Control
System for Maximum Accuracy and Performance**

.

Submitted by:

Team Name : BOD_Paradox

Team Members:

Shadman Sarwer 22201242

H. M. Tahsin Sheikh 22201243

Humayra Jihan Arpita 22201244

Section : E-1

Submitted to:

A S Zaforullah Momtaz

Assistant Professor,

Department of Computer Science and Engineering

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Project Goals:

- To design and develop a high-speed autonomous line-following robot.
- To implement a custom 6-channel IR sensor array for accurate line detection.
- To apply Calibration control for accurate and stable robot movement.
- To improve speed while maintaining line tracking accuracy.
- To gain practical knowledge of robotics, embedded systems, and automation.
- To understand the integration of sensors, motor drivers, and microcontrollers in real-world robotic systems.

Introduction

A line-following robot is an autonomous robot that follows a predefined path using infrared (IR) sensors without human control. The robot detects the line position and continuously adjusts its movement to stay on track.

In this project, we developed a high-speed line-following robot using a custom 6-channel IR sensor array and a calibrated PID (Proportional–Integral–Derivative) control algorithm. Unlike basic robots that use simple movement logic, our robot uses continuous sensor feedback for smoother movement, better stability, and more accurate turning on curved tracks.

The 6-channel sensor array improves line detection accuracy, while the PID controller helps minimize errors and maintain precise steering. This project combines concepts from embedded systems, electronics, control systems, and robotics, and demonstrates real-world applications of autonomous navigation in industries, warehouses, and automated transportation systems.

Social Impact and Motivation

Automation has become an essential part of modern industries and smart systems. This project demonstrates how autonomous robotic systems can improve efficiency, reduce human effort, and increase operational accuracy. Line-following robots are widely used in factories, warehouses, hospitals, and automated production systems for material transportation and repetitive tasks. These robots reduce the need for manual labor in risky or physically demanding environments.

Social Impact

- Improves industrial automation efficiency.
- Reduces repetitive human labor.
- Minimizes operational errors in transportation systems.
- Enhances safety in industrial environments.
- Encourages technological learning among students.

Motivation

Our main motivation behind this project was to gain practical experience in robotics and autonomous systems. We wanted to understand how sensors, controllers, and motor systems interact to perform intelligent movement. Additionally, we aimed to develop a robot capable of maintaining high speed while preserving stability and accuracy using Calibration control. This project also helped us improve our teamwork, programming, circuit design, and hardware integration skills.

Required Modules:

Components Used:

1. Controller:
 - Arduino Nano.
 - Main microcontroller used for processing sensor data and controlling motors
2. Sensors:
 - Custom 6-Channel IR Sensor Array
 - Detects the line position accurately
3. Motors:
 - 2 DC Gear Motors
 - Provides movement to the robot
4. Motor Driver:
 - L298N or L293D
 - Controls motor direction and speed
5. Mechanical Parts:
 - Chassis (Main body structure)
 - Wheels and Castor wheel(Supports movement and balance)
6. Power Supply:
 - Lithium Battery
7. Wires:
 - Jumper Wires and Connectors.

Circuit Diagram:

The robot circuit consists of an Arduino Nano connected with a custom 6-channel IR sensor array and an L298N motor driver module. The IR sensors continuously detect the line position and send data to the microcontroller. Based on the sensor readings, the Arduino calculates the correction using the Calibration algorithm and controls the motors accordingly.

The motor driver receives PWM signals from the Arduino and adjusts the motor speed and direction.

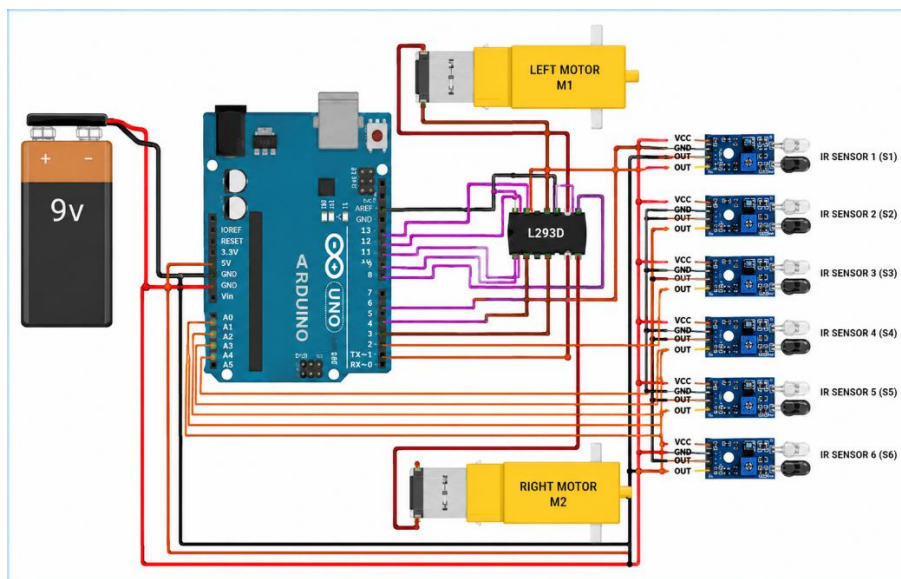


Figure 1: Line Following Robot circuit diagram

Circuit Design Tools:

- Tinker CAD.

Project Development Methodology

The development of this project was completed through a series of systematic and well-planned steps to ensure better performance, stability, and accuracy of the line-following robot.

Step 1: Understanding the Problem

First, we studied the basic working principles of autonomous line-following robots and researched how IR sensors and PID calibration techniques are used in robotic navigation systems. This helped us understand how robots detect paths and make real-time movement decisions.

Step 2: Requirement Analysis

We analyzed the main project requirements such as robot speed, line detection accuracy, movement stability, sensor coverage, power efficiency, and response time. These factors were important for designing a fast and reliable robot suitable for competition tracks.

Step 3: Component Selection

Components were selected carefully based on performance, availability, durability, and cost efficiency. The main components included an Arduino Nano, IR sensors, DC geared motors, motor driver module, rechargeable battery, and robot chassis.

Step 4: Sensor Array Design

A custom-built 6-channel IR sensor array was developed to improve line detection capability. Compared to traditional 2 or 3 sensor systems, the 6-channel array provided wider sensing coverage and more accurate positional feedback, especially during sharp turns and curved paths.

Step 5: Circuit Design

The complete electrical circuit was designed and verified using simulation software such as TinkerCAD. Proper power distribution and sensor connections were checked before hardware implementation to reduce errors during assembly.

Step 6: Hardware Assembly

All hardware components including the sensors, motors, motor driver, controller board, wheels, caster ball, and battery pack were assembled on the robot chassis. The sensors were positioned carefully to maintain proper distance from the track surface.

Step 7: Programming and PID Calibration Implementation

The Arduino Nano was programmed using the Arduino IDE. A calibrated PID (Proportional–Integral–Derivative) control algorithm was implemented to achieve smooth and stable movement. The PID controller continuously corrected steering errors based on sensor feedback, allowing the robot to move efficiently even at higher speeds.

Step 8: Sensor Calibration

The IR sensors were calibrated on both black and white surfaces to determine accurate threshold values. This calibration improved line detection reliability and minimized incorrect sensor readings caused by lighting conditions.

Step 9: Testing and Debugging

The robot was tested on different types of tracks including straight lines, curves, and sharp turns. Problems related to motor speed imbalance, delayed turning response, and sensor noise were identified and corrected through repeated debugging.

Step 10: Performance Optimization

Several parameters such as motor speed, PID constants, and sensor thresholds were optimized to improve tracking performance. This process reduced movement instability, improved corner handling, and increased overall robot speed and accuracy.

Step 11: Final Implementation

After successful testing and optimization, the final robot was prepared for demonstration and competition. The completed robot was capable of following lines smoothly, handling curves accurately, and maintaining stable movement with minimal tracking error. The project successfully demonstrated the practical application of embedded systems, robotics, autonomous navigation, and control algorithms in real-world engineering solutions.

Implementation / Working Procedure

The working procedure of the robot is based on continuous line detection and automatic motor control.

Sensor Detection

The custom 6-channel IR sensor array continuously detects the position of the black line on the white surface. Each sensor sends digital signals to the Arduino Nano depending on whether the surface is black or white.

Data Processing

The Arduino Nano reads all sensor values and determines the current position of the robot relative to the line.

Calibration Control Logic

A Calibration (Proportional-Integral-Derivative) control algorithm is used to calculate the error between the robot position and the center of the line. Based on this error value, correction signals are generated.

- The Calibration controller helps:
- Reduce movement instability
- Improve turning smoothness
- Maintain stability at high speed
- Increase line tracking accuracy

Motor Control

The correction value generated by the Calibration controller is sent to the L298N motor driver. The motor driver adjusts the speed of the left and right motors independently.

- If the robot moves left from the line, the controller increases the speed of one motor and decreases the other.
- If the robot moves right, the opposite correction is applied.
- If the robot is centered, both motors move at equal speed.

Autonomous Navigation

Using continuous feedback from the sensors and Calibration control, the robot can autonomously follow straight paths, curves, and sharp turns while maintaining stable movement.

Testing and Results

Several tests were conducted to evaluate the performance of the robot.

Straight Line Test

The robot successfully followed straight tracks with stable movement and minimal deviation.

Curved Path Test

The Calibration controller improved turning performance and reduced overshooting during curved path navigation.

High-Speed Performance Test

The robot-maintained line tracking accuracy even at higher speeds compared to traditional line-following robots.

Sensor Accuracy Test

The custom 6-channel IR sensor array provided accurate line detection and better positional feedback.

Challenges Faced

During testing, several challenges were identified:

- Sensor sensitivity variation
- Oscillation at high speed
- Sudden deviation during sharp turns
- Battery voltage fluctuation
- Solutions Applied
- Sensor calibration was performed repeatedly.

- Calibration constants were tuned for better stability.
- Motor speed balancing was adjusted.
- Mechanical alignment was improved.

Final Result

After optimization, the robot achieved smoother movement, stable turning, and accurate high-speed line following.

Budget Preparation of the Project:

Serial	Component	Quantity	Unit Price	Total
1	Arduino Nano	1	400	400
2	IR Sensors	6 arrays	350	350
3	Motor Driver	1	200	200
4	DC Motors	2	300	300
5	Chassis	1	400	400
6	Battery	2	300	300
7	Miscellaneous	1	300	800
	Total Estimated Cost			2,750₳ (Two thousand seven hundred fifty Taka.)

Table 1: Budget Table

CEP Mapping of the Project

1. Knowledge Profile (K's) Mapping

K's Attributes How K's Are Addressed Through Our Project CO PO				
K Code	Attribute	How K's Are Addressed Through Our Project	Related COs	Related POs
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline	In this project, we applied basic engineering concepts such as Ohm's Law, voltage-current relationship, and digital logic. The working of IR sensors, motor drivers, and microcontroller programming is based on fundamental electronics and embedded system theory.	CO1, CO2	PO1
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline	We used knowledge of embedded systems, sensor interfacing, and motor control to build the robot. Understanding how IR sensors detect lines and how motor drivers control movement shows the use of specialized robotics knowledge.	CO2, CO3	PO2, PO3
K5	Knowledge that supports engineering design in a practice area	The robot was designed by selecting suitable components like sensors, motors, and controller. We developed the control logic to ensure proper movement and path following. This shows practical design knowledge.	CO3, CO4	PO3, PO5,
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	We implemented the project using real hardware such as Arduino, sensors, and motors. We followed proper circuit connections, testing methods, and debugging techniques, which reflect practical engineering practice.	CO4, CO5	PO4, PO5

2. Complex Engineering Problems (P's) Mapping

P's Attributes How P's Are Addressed Through Our Project CO PO				
P Code	Attribute	How P's Are Addressed Through Our Project	Related COs	Related POs
P1	Depth of knowledge required	The project requires understanding of multiple areas like electronics, programming, and control systems. Combining these concepts to make the robot work properly shows depth of knowledge.	CO1, CO2	PO1
P2	Range of conflicting requirements	We had to balance speed and accuracy. If the robot moves too fast, it may miss the line. If too slow, performance decreases. So, we adjusted motor speed to solve this issue.	CO2, CO3	PO2, PO3
P7	Interdependence	The system depends on multiple components working together. Sensors, controller, and motors must work properly at the same time. If one part fails, the whole system is affected.	CO3, CO4	PO4

3. Complex Engineering Activities (A's) Mapping

A Code	Attribute	How A's Are Addressed Through Our Project
A1	Range of resources	We used different resources such as hardware components (Arduino, sensors, motors), software tools (Arduino IDE), and online learning materials.
A2	Level of interaction	The project required teamwork. Group members worked together in designing, coding, assembling, and testing the robot.
A5	Familiarity	Although line following robots are common, we applied our own understanding to build and test the system. This helped us gain familiarity with real engineering work.

Conclusion

This project successfully demonstrated the design and implementation of a high-speed autonomous line-following robot using a custom 6-channel IR sensor array and Calibration control. Through this project, we gained practical experience in robotics, embedded systems, sensor calibration, motor control, and autonomous navigation. We also learned how Calibration control improves movement stability and accuracy compared to basic line-following techniques. During development, several challenges were encountered including sensor sensitivity issues, speed balancing, and turning stability. Through repeated testing and optimization, these problems were minimized and the robot performance was significantly improved. This project also highlighted the importance of automation in modern industries. Similar robotic systems are widely used in industrial transportation, warehouse automation, and smart manufacturing systems. Overall, this project enhanced our technical skills, problem-solving abilities, teamwork, and interest in advanced robotics and intelligent autonomous systems.

Appendix

Survey 1

We interviewed the president of our robotics club to gather practical insights and expert opinions regarding line following robots (LFR). During the discussion, he explained that line following robots are considered one of the most fundamental yet highly important projects in the field of robotics. According to him, working on LFR projects helps students build a strong foundation in embedded systems, sensor integration, motor control. He also mentioned that these projects provide valuable hands-on experience in programming, circuit design, and problem-solving, which are essential skills for robotics and automation engineering. Through designing and testing a line following robot, students learn how sensors collect data from the environment and how microcontrollers process that data to make real-time decisions.



Figure 1: Interview with Robotics Club President



Figure 2: Attending Workshop

Survey 2

Participating in the Line Following Robot Competition at the Robo Expo 3.0 was a great opportunity to showcase our robotics and programming skills. The expo gathered many innovative teams and created a competitive environment where students demonstrated autonomous robots capable of solving real-world tasks through automation and intelligent control systems. Robotics competitions encourage hands-on learning, innovation, and practical engineering experience. To become a winner, our main focus was speed, accuracy, and stability. We calibrated the IR sensors carefully, optimized motor speed, reduced response delay, and improved turning control for sharp corners. During the competition, the robot successfully maintained line tracking with minimal deviation, completed the track efficiently, and demonstrated reliable autonomous navigation. This performance highlighted our teamwork, coding skills, and practical robotics knowledge.



Figure 3: Line Following Robot Competition Participation in Robo Expo 3.0



Figure 4: During LFR Competition

Team Working Photos

Working as a team in the Line Following Robot (LFR) project helped us improve collaboration, communication, and problem-solving skills. Each team member contributed to different parts of the project such as circuit design, sensor setup, programming, chassis assembly, and testing. Through teamwork, we were able to identify errors quickly, improve robot performance, and complete the project efficiently. Team coordination also helped us handle competition pressure and develop innovative solutions for better line tracking and speed control. The experience strengthened our technical knowledge as well as our ability to work together toward a common goal.



Figure 5: Fixing the Tires and Continuing the Assembly

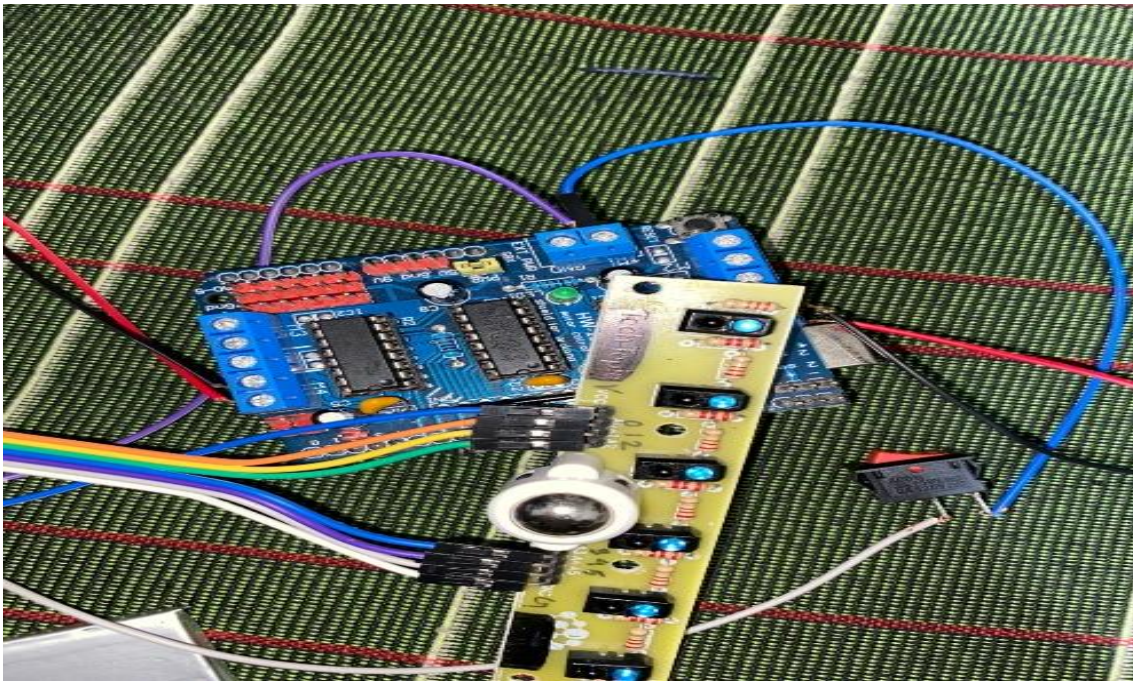


Figure 6: Assembling the IR sensor to the motor driver

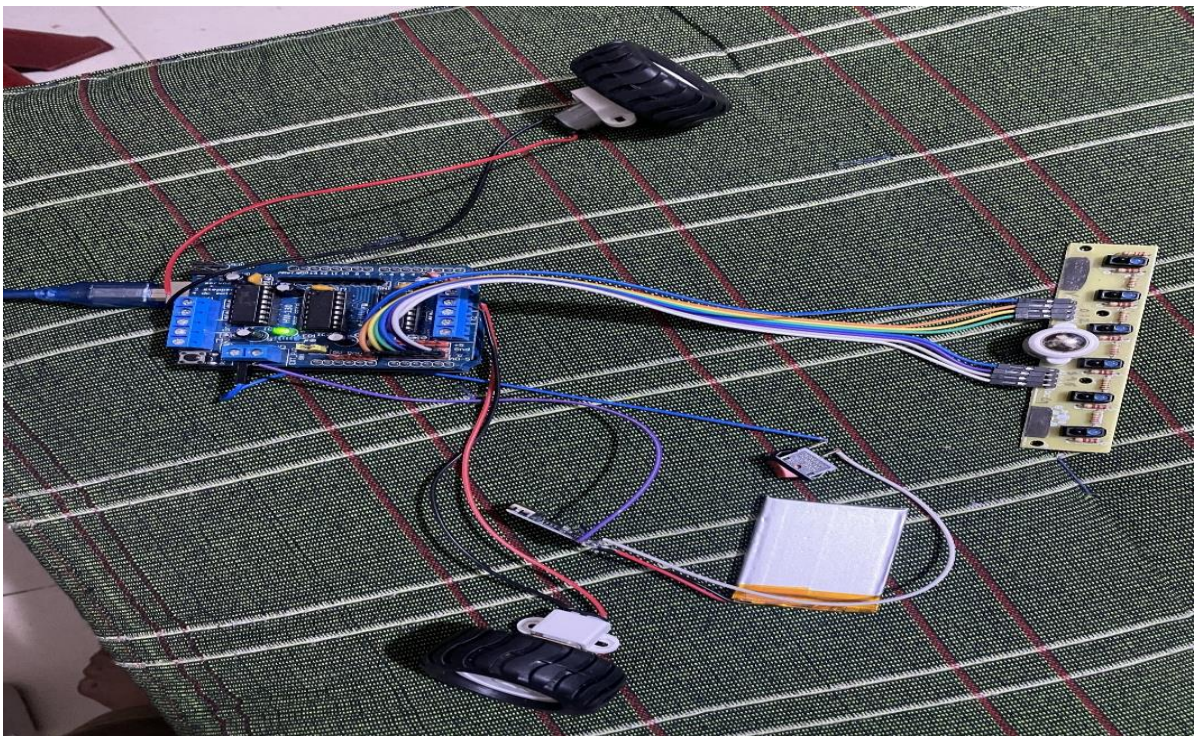


Figure 7: Assembling the motor to the motor driver

Final Robot Images

Our robot was designed using 6 IR sensors, an Arduino controller, and two DC motors. The IR sensors continuously detected the black line on the white track and sent signals to the microcontroller. Based on the sensor readings, the robot automatically adjusted motor speed and direction to follow the path smoothly and accurately. The weighted sensor algorithm helped the robot make fast decisions during curves and sharp turns, improving stability and reducing tracking error.

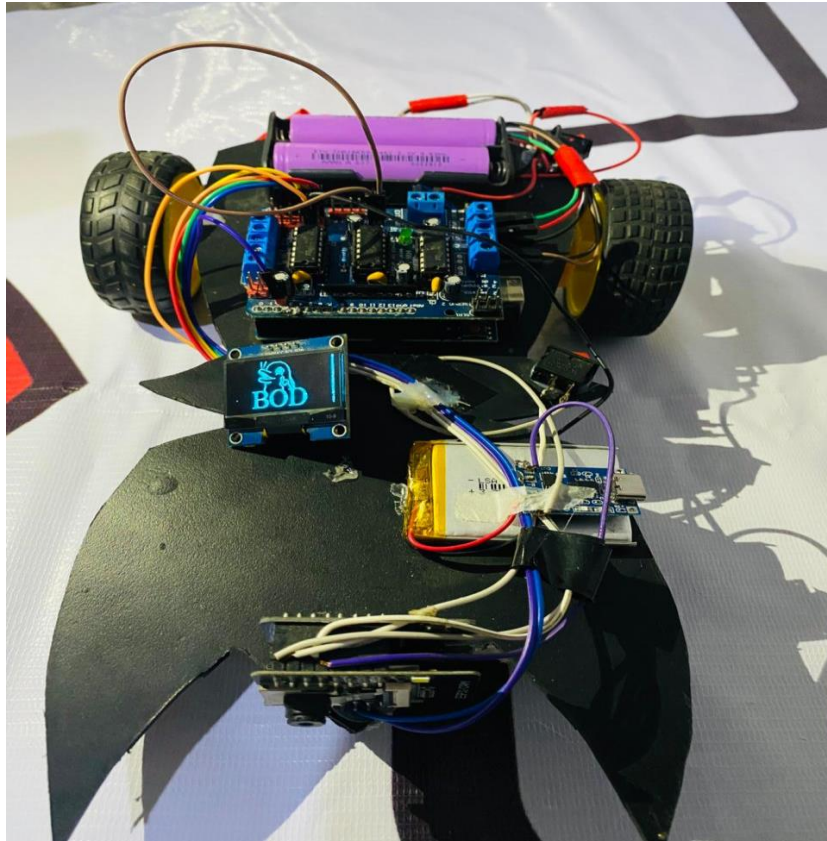


Figure 8: Line Following Robot (BOD_Paradox)

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